

Solvency Field Theory V13.6

From Quantum Settlement Rules
to Atomic Structure and Chemical Bonding

Narrative companion to the
V13.6 Quantum-to-Chemistry Mechanism Stack

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The same settlement mechanics that steer a particle's next write also build the first skeleton of chemistry.

Publication boundary. V13.6 derives uncertainty, tunneling, Pauli exclusion, electron shell capacities, the Aufbau filling skeleton, and the first covalent bond from native settlement mechanics. All results are structural kernels — the qualitative mechanisms are earned, the quantitative precision is open. Full atomic spectroscopy, exact bond energies, Hund's rules, and valence chemistry remain future targets.

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1 Why particles are uncertain

The particle orbits inside its own settlement well at $r_{\text{spin}} = \hbar/(2mc)$. The orbit follows the local settlement geometry. That geometry includes contributions from every other particle and photon whose settlement extends to this location.

The particle can model its own settlement. It cannot model everyone else's. The current geometry at the next write location includes unknowable contributions from everything nearby — other particles oscillating, photons transiting, the entire accumulated write history of the universe at that point. The particle's trajectory is determined by the total geometry but unpredictable because the total geometry is computationally irreducible from inside.

The effective diffusion comes from the closure scale:

$$\boxed{D_{\text{eff}} = c \times r_{\text{spin}} = \frac{\hbar}{2m}}. \quad (1)$$

The write rate is c . The unresolvable scale is r_{spin} . The product is the diffusion constant. Heavier particles have smaller orbits, resolve the background better, and diffuse less. Lighter particles have larger orbits and diffuse more.

The uncertainty principle follows:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (2)$$

Uncertainty is not randomness added to physics. It is the particle's inability to compute the current total settlement geometry at its next write location. The worldline records every dent faithfully. But the next write cannot resolve the current dents before it lands.

2 Why particles tunnel

A potential barrier is a settlement ridge — a region where the local write cost exceeds the particle's available energy. Classically, the particle can't cross. But the settlement support-readiness doesn't stop at the ridge's edge. It extends continuously through the barrier, attenuating exponentially because each layer of barrier material taxes the support:

$$\psi_{\text{far}} \sim e^{-\int \kappa(x) dx}, \quad \kappa(x) = \frac{\sqrt{2m(V(x) - E)}}{\hbar}. \quad (3)$$

The settlement intensity on the far side is:

$$I_{\text{far}} \sim e^{-2 \int \kappa(x) dx}. \quad (4)$$

The V13.5 write-race mechanism applies: the far-side channel has nonzero settlement intensity, so it enters the next-write race with nonzero hazard. Most ticks, a near-side address wins. Rarely, the far side wins. That rare event is tunneling.

Thinner barriers mean less attenuation, more far-side intensity, more frequent tunneling. Higher barriers mean faster attenuation, less far-side intensity, rarer tunneling. The exponential dependence on thickness comes from layer multiplication — each barrier layer taxes the support by a factor less than one, and many layers multiply to give $e^{-\kappa d}$.

Tunneling is not probability magic. It is far-side write completion from closure support that survives exponential taxation by the barrier's settlement ridge.

3 Why two electrons can't share a state

The fermion has 4π spinor closure. A 2π rotation — which is what exchanging two identical particles amounts to geometrically — returns the spinor to minus itself. The settlement amplitude picks up a factor of -1 :

$$\Psi(1, 2) = -\Psi(2, 1). \quad (5)$$

Set both particles in the same complete state:

$$\Psi(q, q) = -\Psi(q, q) = 0. \quad (6)$$

Zero settlement intensity. Zero write-race entries. Zero probability. The same-state address doesn't merely reject the second fermion. The address CEASES TO EXIST for two identical fermions. The settlement geometry itself vanishes at the double-occupation point.

Bosons, with 2π closure, pick up $+1$ under exchange: $\Psi(1, 2) = +\Psi(2, 1)$. Same state is nonzero. No exclusion. Bosons can share.

This is the spin-statistics connection: 4π closure gives antisymmetry gives exclusion. 2π closure gives symmetry gives sharing. From the same closure geometry that produced the mass hierarchy, the gravitational coupling, and the charge spectrum.

4 How atoms build shells

An electron in a central settlement well (around a nucleus) has a complete state labeled by four quantum numbers: (n, ℓ, m_ℓ, m_s) . The radial band n , the angular pattern ℓ , the orientation m_ℓ , and the spin closure direction $m_s = \pm \frac{1}{2}$.

Pauli exclusion says no two electrons can share all four. The counting follows:

Each ℓ has $2\ell + 1$ orientation addresses. Each can hold two spin directions. Subshell capacity: $2(2\ell + 1)$. So $s = 2$, $p = 6$, $d = 10$, $f = 14$.

Each shell n contains subshells from $\ell = 0$ to $\ell = n-1$. Total shell capacity: $2n^2$. So shell 1 holds 2, shell 2 holds 8, shell 3 holds 18, shell 4 holds 32.

The periodic table's period lengths — 2, 8, 8, 18, 18, 32, 32 — follow from this capacity structure combined with the energy-ordering rule.

5 Why shells fill in the order they do

Each electron settlement address has a cost determined by two components:

Radial band cost n — how many nodes the radial settlement support has. More nodes mean higher kinetic settlement cost. Like a vibrating string with more harmonics.

Angular barrier tax ℓ — how much angular momentum prevents the electron from penetrating close to the nucleus. Higher ℓ means the electron is kept farther from the deep nuclear settlement well. Less binding. Higher effective cost.

The total cost scales as $n + \ell$. Lower total cost fills first. When two addresses have the same $n + \ell$, the lower n fills first because lower- n orbits penetrate deeper into the nuclear well.

This gives the Aufbau ordering:

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow \dots \quad (7)$$

The entire filling order of the periodic table from settlement economics. Not a rule imposed by hand. A consequence of minimizing write cost in a central settlement well with angular barriers.

6 The first chemical bond

Take two hydrogen atoms. Each has one proton creating a $1/r$ directional settlement well and one electron orbiting in the $1s$ ground state. Bring them together.

As the atoms approach, their settlement support functions overlap. The region between the two protons receives settlement contributions from BOTH nuclear wells simultaneously. The settlement intensity between the protons is enhanced — deeper than either well alone.

The electrons notice. Their write races now include addresses in the shared settlement basin between the protons. The shared basin has higher intensity than the individual atomic basins. The electrons preferentially complete there. Both electrons migrate toward the shared region.

Pauli exclusion allows this: the two electrons can share the same spatial bonding basin because they have opposite spin closures ($m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$). Different complete settlement addresses. No exclusion.

The bonding state $\psi_+ \sim \psi_A + \psi_B$ is a settlement bridge — enhanced intensity between the protons. The antibonding state $\psi_- \sim \psi_A - \psi_B$ is a settlement cut — a node between the protons where the intensity drops to zero.

The system is more stable in the bonding state because the shared settlement basin lowers the total write cost below the separated atoms. The energy difference IS the bond energy. The equilibrium separation IS the bond length. The molecule H_2 exists because shared settlement is cheaper than separate settlement.

SM correspondence: molecular orbital theory; LCAO approximation; exact bond energy 4.5 eV and bond length 74 pm remain open targets.

7 What this means

From energy at c to chemistry in six steps:

1. Uncertainty: the particle can't resolve the current settlement geometry. $D_{\text{eff}} = \hbar/(2m)$.
2. Tunneling: settlement support extends through barriers. Far-side write completion at exponentially reduced intensity.
3. Exclusion: 4π closure means exchange flips sign. Same-state amplitude vanishes.
4. Shells: exclusion plus central settlement gives $2(2\ell + 1)$ per subshell, $2n^2$ per shell.
5. Ordering: radial cost plus angular tax gives the $n + \ell$ Aufbau rule.
6. Bonding: overlapping settlement support creates a shared basin. Molecule is cheaper than separate atoms.

Each step uses the same settlement mechanics. Each step builds on the previous. Each step was derived from structures that were established for other purposes — the 4π closure that gave the mass hierarchy also gives the periodic table. The settlement intensity that gives the Born rule also gives the bond.

The universe doesn't need different rules at different scales. It needs one rule — no-overwrite — operating on one stuff — energy at c — and the hierarchy of structure from particles through atoms through molecules through chemistry emerges from settlement mechanics applied at each successive scale.

The V13 series

V13.0 — particles.

V13.1 — gravity.

V13.2 — expansion.

V13.3 — the first atom.

V13.4 — forces.

V13.5 — ontology and entanglement.

V13.6 — quantum rules, atomic structure, and the first chemical bond.

Seven chapters. One mechanism. From energy at c to H_2 .

The pen is still writing. The document is still growing. And the next page is chemistry.

Acknowledgments

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